

LONGKOU, China

WMO: 547530

Lat: 37.6394N

Lon: 120.3359E

Elev: 5

StdP: 101.26

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-6.1	-16.1	0.9	-2.8	-14.8	1.0	-2.3	8.8	-2.4	8.0	-2.4	2.5	180	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	32.9	24.8	31.7	24.2	30.5	23.8	27.0	31.1	26.2	29.9	25.5	28.9	3.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	29.8	25.1	20.2	28.9	24.5	19.5	28.3	85.2	31.2	82.0	29.9	78.9	29.0	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.4	6.6	-9.3	35.7	1.5	1.7	-10.4	36.8	-11.3	37.8	-12.2	38.8	-13.3	40.0		
			-10.4	28.1	1.5	0.6	-11.4	28.5	-12.3	28.9	-13.1	29.3	-14.2	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	-1.1	0.7	6.2	13.1	19.3	23.6	26.5	25.9	21.7	15.6	8.2	1.7
	DBStd	10.22	2.79	3.54	4.35	4.17	3.51	2.74	2.36	2.27	2.56	3.19	3.79	2.93
	HDD10.0	1094	346	260	134	15	0	0	0	0	0	2	80	258
	HDD18.3	2587	604	493	376	163	32	1	0	0	4	94	306	516
	CDD10.0	2379	0	0	17	109	287	408	512	493	351	177	24	0
	CDD18.3	830	0	0	1	7	60	159	254	234	104	11	0	0
	CDH23.3	6853	0	0	3	43	394	1203	2514	2128	540	28	0	0
	CDH26.7	2072	0	0	0	7	97	366	882	637	84	1	0	0
Wind	WSAvg	3.2	3.1	3.2	3.6	3.9	3.6	3.3	3.0	2.7	2.7	3.0	3.4	3.3
Precipitation	PrecAvg	618	8	10	13	33	42	71	171	138	64	35	23	10
	PrecMax	945	39	38	75	108	134	257	370	308	174	88	96	26
	PrecMin	402	0	0	1	3	1	4	41	28	3	2	0	0
	PrecStd	143	10	10	15	27	34	62	81	77	44	26	21	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	13.2	22.1	27.2	31.8	33.9	34.9	33.7	30.7	25.8	19.9	12.3
		MCWB	4.0	7.4	12.8	16.5	19.7	21.3	26.8	26.9	23.3	17.4	13.5	8.8
	2%	DB	6.4	10.1	18.0	24.3	29.2	31.8	33.3	32.2	28.7	24.0	17.6	9.5
		MCWB	2.9	5.0	10.5	15.0	18.4	21.4	25.8	25.9	21.5	17.1	12.1	6.1
	5%	DB	4.7	8.1	15.3	22.1	27.2	30.3	32.0	30.9	27.5	22.3	15.8	7.6
		MCWB	1.7	3.9	8.5	13.6	17.2	20.9	25.2	25.3	21.2	16.3	11.4	4.5
	10%	DB	3.2	6.3	12.9	20.0	25.5	28.8	30.7	29.8	26.2	20.7	14.2	6.2
		MCWB	0.5	2.7	7.1	12.5	16.5	20.5	24.8	24.7	20.4	15.7	10.2	3.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	8.2	13.2	18.1	21.2	24.9	28.0	28.0	25.2	19.9	15.0	9.2
		MCDB	7.4	12.0	20.4	24.8	29.2	30.3	33.1	32.5	28.1	23.4	17.5	11.2
	2%	WB	3.4	5.8	11.1	16.4	19.8	23.6	27.1	27.0	23.6	18.7	13.5	6.7
		MCDB	5.8	9.4	16.9	22.5	27.0	28.6	31.7	30.8	27.0	21.9	16.2	9.0
	5%	WB	2.1	4.4	9.2	14.7	18.7	22.5	26.4	26.3	22.5	17.6	12.0	5.0
		MCDB	4.3	7.3	14.2	20.6	24.9	27.3	30.5	29.7	25.9	21.0	15.3	7.0
	10%	WB	0.7	3.1	7.5	13.3	17.7	21.6	25.7	25.7	21.6	16.5	10.5	3.6
		MCDB	2.8	6.0	12.2	18.7	23.5	26.5	29.4	28.8	25.0	20.0	13.8	5.9
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.7	7.8	8.4	8.4	7.8	6.3	6.0	7.1	7.2	6.6	5.6
		MCDBR	8.5	9.6	11.1	11.4	11.0	10.1	8.0	7.0	8.2	9.0	9.0	7.9
		MCWBR	6.2	6.3	6.5	6.1	5.4	4.3	3.3	3.1	3.9	4.7	5.8	5.9
	5% WB	MCDBR	7.4	8.4	10.1	10.5	9.5	7.8	6.8	6.1	6.7	7.5	7.8	7.1
		MCWBR	5.9	6.0	6.4	6.2	5.3	3.9	3.4	3.1	3.8	4.8	6.0	5.8
Clear-Sky Solar Irradiance	taub		0.490	0.546	0.589	0.623	0.651	0.753	0.739	0.671	0.592	0.614	0.562	0.505
	taud		1.812	1.724	1.667	1.653	1.639	1.513	1.591	1.703	1.799	1.674	1.736	1.810
	Ebn at Noon		632	654	677	688	679	612	616	646	669	586	557	580
	Edh at Noon		160	198	229	245	253	287	264	231	198	205	172	151
All-Sky Solar Radiation	RadAvg		2.23	3.07	4.38	5.19	5.75	5.42	4.77	4.63	4.24	3.30	2.34	1.93
	RadStd		0.21	0.28	0.33	0.22	0.45	0.42	0.48	0.46	0.33	0.29	0.30	0.15

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+77	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+74	N/A

Nomenclature: See separate page